

1 Logic

1.1 Basics

Lemma (2.1)

- $A \wedge A \equiv A$ and $A \vee A \equiv A$ (**Idempotence**)
- $A \wedge B \equiv B \wedge A$ and $A \vee B \equiv B \vee A$ (**Commutativity**)
- $(A \wedge B) \wedge C \equiv A \wedge (B \wedge C)$ and $(A \vee B) \vee C \equiv A \vee (B \vee C)$ (**Associativity**)
- $A \wedge (A \vee B) \equiv A$ and $A \vee (A \wedge B) \equiv A$ (**Absorption**)
- $A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$ (**1st Distributive**)
- $A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$ (**2nd Distributive**)
- $\neg \neg A \equiv A$ (**Double Negation**)
- $\neg(A \wedge B) \equiv \neg A \vee \neg B$ and $\neg(A \vee B) \equiv \neg A \wedge \neg B$ (**de Morgan**)

Logical Consequence $F \models G$ if G true whenever F true.

Equivalence $F \equiv G \iff F \models G$ and $G \models F$

Alt. Def. Tautology $\models F$ is equivalent to $\top$.

Satisfiability F satisfiable if true for at least one truth assignment.

Lemma (2.3) $(F \rightarrow G \equiv \top) \iff (F \models G)$

1.2 Predicates

A k -ary predicate P on U is a function $U^k \rightarrow \{0, 1\}$, i.e., operates on k elements.

Short-hand: $\forall x \geq 5 : F$ instead of $\forall x (x \geq 5 \rightarrow F)$

1.3 Proof Patterns

For proving $S \implies T$:

- Composition of Implications
- Direct Proof (assume S , prove T)
- Indirect Proof (assume $\neg T$, prove $\neg S$)
- Modus Ponens (find statement R , prove R , prove $R \implies S$)
- Case Distinction
- Proof by Contradiction (find statement T , prove $\neg T$; assume S false and prove T to reach contradiction)
- Existence Proof, Proof by Counterexample, Induction
- Pigeonhole Proof (if a set of n objects is partitioned into $k < n$ sets, at least one contains $\lceil \frac{n}{k} \rceil$ objects – e.g., “one set contains at least two of this”)

2 Sets

Equality $A = B \iff \forall x (x \in A \leftrightarrow x \in B)$ or $(A \subseteq B) \wedge (B \subseteq A)$

Element Equality $\{a\} = \{b\} \implies a = b$

Subset $A \subseteq B \iff \forall x (x \in A \rightarrow x \in B)$ [**Transitive per L 3.3**]

Union $A \cup B = \{x \mid x \in A \vee x \in B\}$

- On a set of sets: $\cup A = \{x \mid x \in a \text{ for some } a \in A\}$

Intersection $A \cap B = \{x \mid x \in A \wedge x \in B\}$

- On a set of sets: $\cap A = \{x \mid x \in a \text{ for all } a \in A\}$

Difference $B \setminus A = \{x \in B \mid x \notin A\}$

Power Set $P(A) = \{S \mid S \subseteq A\}$ – cardinality $2^{|A|}$

Theorem (3.4 - Set Laws) $\forall A, B, C$:

Idempotence: $A \cap A = A$ and eq for $\cup$

Commutativity: $A \cap B = B \cap A$ and eq for $\cup$

Associativity: $A \cap (B \cap C) = (A \cap B) \cap C$ and eq for $\cup$

Absorption: $A \cap (A \cup B) = A$ and $A \cup (A \cap B) = A$

Distributivity: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Consistency: $A \subseteq B \iff A \cap B = A \iff A \cup B = B$

Cartesian Product $A \times B$; Empty set if either $A, B = \emptyset$

Example – Definition of $a \equiv b \pmod m$: $a - b = km$ for some k . i.e., $m \mid (a - b)$.

Identity $\text{id}_A = \{(a, a) \mid a \in A\}$

Inverse $a\hat{\rho}b = b\rho a$ — for composition, $\widehat{\rho \circ \sigma} = \hat{\sigma} \circ \hat{\rho}$

Composition $\rho \circ \sigma = \{(a, c) \mid \exists b (a\rho b \wedge b\sigma c)\}$ (**Associative**)

Transitive Closure $\rho^+ = \cup_{n \in \mathbb{N} \setminus \{0\}} \rho^n$ (for transitive ρ : $\rho^n \subseteq \rho$)

2.1.1 Types

Name	Formula	Set	Example
Reflexive	$a\rho a$	$\text{id} \subseteq \rho$	$\text{id}, \geq$
Irreflexive	$\neg(a\rho a)$	$\text{id} \cap \rho = \emptyset$	$\neq, >$
Symmetric	$a\rho b \iff b\rho a$	$\rho = \hat{\rho}$	$\text{id}, \equiv \pmod m$
Antisymmetric	$a\rho b \wedge b\rho a \rightarrow a = b$	$\rho \cap \hat{\rho} \subseteq \text{id}$	$\text{id}, \geq, $
Transitive	$a\rho b \wedge b\rho c \rightarrow a\rho c$	$\rho^2 \subseteq \rho$	$\text{id}, \equiv \pmod m, >$

Equivalence Relation Reflexive, Symmetric, Transitive (e.g., $\equiv \pmod m$ on $\mathbb{Z}$). The intersection of two equivalence relations is itself one.

Equivalence Class $[a]_\theta = \{b \in A \mid b\theta a\}$

- Complete relation $A \times A$: $\forall a, b : [a] = [b]$;
- id where $\forall a, b : [a] \neq [b]$

Classes form a partition: $S_i \cap S_j = \emptyset$ where $i \neq j$. $\cup_{i \in I} S_i = A$

The set of equivalence classes A/θ of a relation $\theta = \{[a]_\theta \mid a \in A\}$ is a partition of A .

Example – Definition of Rationals $(a, b) \sim (c, d) \iff ad = bc$, whereby an equivalence class consists of all the tuples simplifying to the same fraction.

2.2 Partial Order Relations

A partial order $\preceq$ on A is reflexive, antisymmetric, and transitive. Denoted $(A; \preceq)$. $\geq$ is a partial order on $\mathbb{Z}$. $>$ is not (not reflexive).

e.g., $|$ is a partial order on $\mathbb{N} \setminus \{0\}$.

Comparability $a, b \in A$ are comparable if $a \preceq b$ or $b \preceq a$. Otherwise incomparable.

Totally ordered by $\preceq$ if $\forall a, b$ comparable.

Well-ordered if totally ordered and every non-empty subset of A has a least element.

Direct Product $(A; \preceq) \times (B; \sqsubseteq)$ is the poset $(A \times B; \preceq)$ defined by:

$$(a_1, b_1) \preceq (a_2, b_2) \iff a_1 \preceq a_2 \wedge b_1 \sqsubseteq b_2$$

Lex Order The partial order $\left(A \times B; \preceq_{\text{lex}}\right)$ is defined as:

$$(a_1, b_1) \preceq_{\text{lex}} (a_2, b_2) \iff a_1 \prec a_2 \vee (a_1 = a_2 \wedge b_1 \sqsubseteq b_2)$$

Definitions $(A; \preceq)$ poset, $S \subseteq A$.

- $a \in A$ is a minimal (maximal) element of A if there is no $b \in A$ with $b \prec a$. ($a \prec b$)
- $a \in A$ is the least (greatest) element of A if $\forall b \in A : a \preceq b$. ($b \preceq a$)
- $a \in A$ is a lower (upper) bound of S if $\forall b \in S : a \preceq b$. ($b \preceq a$)
- $a \in A$ is the greatest lower bound (least upper bound) of S if a is the greatest (least) element of the set of all lower (upper) bounds of S .

Example 3.47. Consider the poset $(\{2, 3, 4, 5, 6, 7, 8, 9\}; |)$ shown in Figure 3.1. It has no least or greatest elements, but 2, 3, 5, and 7 are minimal elements, and 5, 6, 7, 8 and 9 are maximal elements. The number 2 is a lower bound (actually the greatest lower bound) of the subset $\{4, 6, 8\}$, and the subset $\{4, 9\}$ has no lower (nor upper) bound.

2.2.1 Hasse Diagrams

Cover a covers $b \iff b \prec a$ and no c between a, b .

Directed graph where vertices are $a \in A$ and edge $(a, b) \iff a$ covers b .

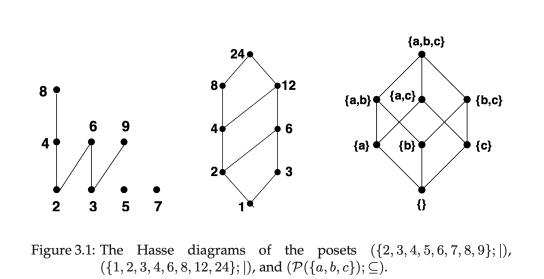


Figure 3.1: The Hasse diagrams of the posets $(\{2, 3, 4, 5, 6, 7, 8, 9\}; |)$, $(\{1, 2, 3, 4, 6, 8, 12, 24\}; |)$, and $(\mathcal{P}(\{a, b, c\}); \subseteq)$.

2.2.2 Meet, Join, Lattices

$(A; \preceq)$:

If $a, b \in A$ have a glb, it is the meet of a, b , denoted $a \wedge b$.

If a, b have lub, it is the join of a, b ($a \vee b$).

If every pair of elements has a meet and join, the poset is called a lattice – e.g., $(\mathbb{N}; \leq)$.

2.3 Functions

$f : A \rightarrow B, f : a \mapsto a^2$ (domain A , codomain B , outputs a^2) is a relation from A to B with:

- $\forall a \in A, \exists b \in B : afb$ (totally defined; otherwise partial function)
- $\forall a \in A, \forall b, b' \in B : (afb \wedge afb' \rightarrow b = b')$ (well-defined)

Thm: If there are injective $f : A \rightarrow B, g : B \rightarrow A$, there is a bijective $h : A \rightarrow B$.

The set of all functions $A \rightarrow B$ is denoted B^A .

A function has a left inverse $\iff$ injective, right inverse $\iff$ surjective.

Image The image of $f : A \rightarrow B$ and $S \subseteq A$, the image $f(S)$ of S under f is the set $f(S) = \{f(a) \mid a \in S\}$. In other words, the output values for a given subset.

Preimage $f^{-1}(T) = \{a \in A \mid f(a) \in T\}$ (set of values in A mapping into T)

Definitions

- Injective (one-to-one) if for $a \neq a'$: $f(a) \neq f(a')$
- Surjective if $f(A) = B$, i.e., for every $b \in B, b = f(a)$ for some $a \in A$ – that is, the codomain is fully mapped to.
- Bijective if both injective and surjective.

Composition $g \circ f$ or $gf = g(f(a))$ (**Associative**)

2.4 Countability

$A \sim B$ if bijection exists. The relation $\sim$ is an equivalence relation. $A \subseteq B \implies A \preceq B$.

$A \preceq B$ if injection $A \rightarrow B$ or $A \sim C$, where C is a subset of B . ($\preceq$ transitive)

Countable $\iff A \preceq \mathbb{N}$

Examples

- Bijection $f : \mathbb{N} \rightarrow \mathbb{Z} - n \mapsto (-1)^n \lceil \frac{n}{2} \rceil$
- $\{0, 1\}^* = \{\epsilon, 0, 1, 00, \dots\}$ countable by prepending 1
- $A \times B$ countable if $A, B \preceq \mathbb{N}$ because $A, B \preceq \{0, 1\}^* \preceq \mathbb{N}$ and bijection $\{0, 1\}^* \times \{0, 1\}^* : (a, b) \mapsto (0^{|a|} \parallel 1 \parallel a \parallel b)$. Hence $\mathbb{Q} \sim \mathbb{N}$.
- $\mathbb{N} \times \mathbb{N}$ countable because $2^* 3^y$ is an injection from $(x, y) \rightarrow \mathbb{N}$ (unique prime factor decomposition)
- $\mathcal{P}(\mathbb{N})$ uncountable per diagonalization. Consider the sets inside the powerset $S_1, S_2, \dots$. Define $f_S(n) = \begin{cases} 1 & \text{if } n \in S \\ 0 & \text{else} \end{cases}$. Then construct a new set that differs from each subset by one element (if S_1 contains 1, this new set won't contain 1, ...). This set will differ from each set within the powerset by at least one element, and hence is not part of the enumeration. So $S_1, S_2, \dots$ is clearly not a full enumeration and thus $\mathcal{P}(\mathbb{N}) > \mathbb{N}$.

Theorem A, A_i countable sets.

$\forall n \in \mathbb{N}, A^n$ (n -tuples – i.e., ordered set of n elements – over A) countable

$\cup_{i \in \mathbb{N}} A_i$ of countable list $A_0, A_1, \dots$ of countable sets countable

A^* of finite sequences of elements from A countable.

$\{0, 1\}^\infty$ **uncountable**.

Proof Assume bijection $f : \mathbb{N} \rightarrow \{0, 1\}^\infty$. Let $\beta_{i,j}$ be the j th bit of the i th bitstring. Let $\bar{\beta}$ be the complement of a given β . Construct a new bitstring $\alpha = \bar{\beta}_{0,0}, \bar{\beta}_{1,1}, \dots$. This bitstring differs by at least one bit from any possible countable bitstring s since $\beta_{s,n} \neq s_n$. So this cannot be a bijection. $\square$

Treat this as binary expansion of a real number in $[0, 1]$ to prove that $\mathbb{R} \geq [0, 1] > \mathbb{N}$.

Function f computable $\iff$ there is a program that outputs $f(n)$ for every $n \in \mathbb{N}$.

3 Number Theory

Thm: Let $f(x_1, \dots, x_k)$ be a multi-variate polynomial in k variables and integer coefficients, $m \geq 1$. Then $R_m(f(a_1, \dots, a_k)) = R_m(f(R_m(a_1), \dots, R_m(a_k)))$.

Theorem (Euclidean Theorem) For all integers a and $d \neq 0$, there are unique q, r such that $a = dq + r$ and $0 \leq r < |d|$ (a dividend, d divisor, q quotient)

3.0.1 Euclidean Algorithm

Start with $m < n$ and replace the pair (m, n) with $R_m(n), m$ until the remainder is zero, at which point the last nonzero number is the gcd of m, n .

3.0.1 Euclidean Algorithm

Compute the Euclidean algorithm step by step

a = 1071; b = 462

1071 = q₀ × 462 + r₀

q₀ = 2; r₀ = 147

462 = q₁ × 147 + r₁

q₁ = 3; r₁ = 21

147 = q₂ × 21 + r₂

q₂ = 7; r₂ = 0

Since r₂ = 0 the algorithm is finished. Thus

GCD(1071, 462) = 21.

Example (RSA) With $p = 61, q = 53$:

- $n = pq = 61 * 53 = 3233$
- $\varphi(n) = 60 * 52 = 3120$
- Choose a coprime e , e.g., $e = 17$.
- Now compute $d \equiv e^{-1} \pmod{3120}$, i.e., $17d \equiv 1 \pmod{3120}$
- $17x + 3120y = 1$
- Euclidean: $3120 = 17 * 183 + 9 \rightarrow 17 = 9 * 1 + 8 \rightarrow 9 = 8 * 1 + 1 \rightarrow 8 = 1 * 8 + 0$ so $\text{gcd}(17, 3120) = 1$

Back substitution:

- From the previous step: $1 = 9 - 8 * 1$
- From the one before: $8 = 17 - 9 * 1$, so:
- $1 = 9 - (17 - 9) = 9 - 17 + 9 = 2 * 9 - 17$
- ...
- $1 = (-367)17 + 2 * 3120 \rightarrow x = -367, y = 2 \rightarrow d = -367 \equiv 2653 \pmod{3120}$

GCD Simplification $\text{gcd}(m, n - qm) = \text{gcd}(m, n)$, thus $\text{gcd}(m, R_m(n)) = \text{gcd}(m, n)$

Ideals $(a, b) = \{ua + vb \mid u, v \in \mathbb{Z}\}$. Every ideal in $\mathbb{Z}$ can be generated by a single integer. Hence: $\text{gcd}(a, b) = ua + vb$

3.1 Expressing lcm, gcd

Per the unique factorization theorem, any $\{a, b\} \subseteq \mathbb{Z}$ can be written as

$$a = \prod_i p_i^{e_i}, b = \prod_i p_i^{f_i}$$
$$\text{gcd}(a, b) = \prod_i p_i^{\min(e_i, f_i)}, \text{lcm}(a, b) = \prod_i p_i^{\max(e_i, f_i)}$$

And because $\min(x, y) + \max(x, y) = x + y$, we get $\text{gcd}(a, b) \cdot \text{lcm}(a, b) = ab$.

3.2 Modulo

See Relations for definition of modular congruence.

$a = b \iff a \equiv b \pmod m$

If $a \equiv b \pmod m, c \equiv d \pmod m$:

$a + c \equiv (b + d) \pmod m$ and $ac \equiv bd \pmod m$

Example 4.14. Remainders can be used to check the correctness of calculations (which were, for instance, carried out by hand). If an error occurred during the computation, it is likely that this error also occurs when the computation is considered modulo some m . To check the result n of a computation one can compare $R_m(n)$ with the remainder modulo m obtained by continuously reducing intermediate results of the computation. The modulus $m = 9$ is especially suited because $R_9(n)$ can be easily computed by adding the decimal digits of n (prove this!), and computing the remainder modulo 9 of this sum. For instance, to check whether $247 \cdot 3158 = 780026$ is correct one can compute $R_9(247) = R_9(2 + 4 + 7) = 4$ and $R_9(3158) = R_9(3 + 1 + 5 + 8) = 8$ to obtain $R_9(247 \cdot 3158) = R_9(4 \cdot 8) = 5$. On the other hand we have $R_9(780026) = R_9(7 + 8 + 2 + 6) = 5$. Hence the result can be correct.

Example 4.15. A similar test can be performed for $m = 11$. $R_{11}(n)$ can be computed by adding the decimal digits of n with alternating sign modulo 11. This test, unlike that for $m = 9$, detects the swapping of digits.

Multiplicative Inverse $ax \equiv 1 \pmod{m}$ has a solution $x \in \mathbb{Z}_m \iff \gcd(a, m) = 1$. Also written $x \equiv a^{-1} \pmod{m}$.

When given u, v with $ua + vb = 1$, $a^{-1} \pmod{m} = R_m(u)$ because $aR_m(u) \equiv au \equiv 1 - vm \equiv 1$.

E.g., compute $142^{-1} \pmod{553}$: Find $\gcd(142, 553)$. We have $553 = 3 \cdot 142 + 127$, then repeat until last step, substituting $(142 = 127 + 15)$ until getting an expression $ua + vb = \gcd(142, 553)$.

3.2.1 Chinese Remainder Theorem

Given pairwise relatively prime $m_1, \dots, m_r$, the system

$$\begin{aligned}x &\equiv a_1 \pmod{m_1} \\&\dots \\x &\equiv a_r \pmod{m_r}\end{aligned}$$

has a unique solution $0 \leq x < \prod_{i=1}^r m_i$, where $0 \leq a_i < m_i$ and $1 \leq i \leq r$.

Application Compute $N = n_1 \dots n_r$. For each congruence, define $N_i = \frac{N}{n_i}$.

Find an M_i such that $N_i M_i \equiv 1 \pmod{n_i}$.

Then sum, $x = \sum_{i=1}^k a_i N_i M_i \pmod{N}$ **and reduce mod N** .

Example Solving $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$:

- $N = 3 \cdot 5 \cdot 7 = 105$
- $N_1 = \frac{105}{3} = 35$, $N_2 = \frac{105}{5} = 21$, $N_3 = \frac{105}{7} = 15$
- $35M_1 \equiv 1 \pmod{3} \rightarrow 2M_1 \equiv 1 \pmod{3} \rightarrow M_1 = 2$, etc.
- $x \equiv 2 \cdot 35 \cdot 2 + \dots \rightarrow x = 23 + 105k_1 \pmod{105}$

Example 4.19. Compute $R_{35}(2^{1000})$. We can do this computation modulo 5 and modulo 7 separately. Since $2^4 \equiv_5 1$ we have $2^{1000} \equiv_5 1$. Since $2^3 \equiv_7 1$ we have $2^{1000} \equiv_7 2$. This yields $2^{1000} \equiv_{35} 16$ since 16 is the (unique) integer $x \in [0, 34]$ with $x \equiv_5 1$ and $x \equiv_7 2$.

3.3 DHKE

- Alice selects a random $x_A \in \{0, \dots, p-2\}$, Bob x_B
- Alice calculates $y_A = R_p(g^{x_A})$, Bob $y_B = R_p(g^{x_B})$
- They exchange y_A, y_B
- Alice calculates $k_{AB} = R_p(y_B^{x_A})$, Bob $k_{BA} = R_p(y_A^{x_B})$

Now, they have $k_{AB} \equiv y_B^{x_A} \equiv (g^{x_B})^{x_A} \equiv g^{x_A x_B} \equiv k_{BA}$

Example Using $\langle 2 \rangle \in (\mathbb{Z}_{15}^\times; \odot)$ ($g = 2$). $y_A = 8$, $k_{AB} = 2$. Find y_B .

$$8 = R_{15}(2^{x_A}) \implies x_A = 3 \implies 2^{3x_B} \equiv_{15} 2 \implies 8^{x_B} \equiv_{15} 2 \implies x_B = 3 \implies y_B = R_{15}(2^3) = 8$$

4 Algebra

Operation An operation on S is a (partial) function $S^n \rightarrow S$ (if $n = 1$, unary; if $n = 2$, binary).

Algebra Written $\langle S; \Omega \rangle$, where S is the carrier and $\Omega = (\omega_1, \dots, \omega_n)$ is a list of operations.

Notation: Redundant operations and elements are sometimes dropped, e.g., $\langle \mathbb{Z}; +, \cdot \rangle$ instead of $\langle \mathbb{Z}; +, -, 0, 1, \cdot \rangle$.

Monoids $\langle M; *, e \rangle$ where $*$ is associative, e neutral. Has one binary and up to one null-, unary operation. Examples being $+$, $*$ on $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and modular $+$, $*$ on $\mathbb{Z}_m$.

Neutral Elements Element $e \in S$ such that $e * a = a$ (left) or $a * e = a$ (right). If there are two neutral elements, they are equal.

Inverses An $\hat{a} \in S$ such that $\hat{a} * a = e$ (left) or $a * \hat{a} = e$ (right). If there are two inverses, they are equal.

Group $\langle G; *, \hat{\cdot}, e \rangle$ satisfying the axioms G1, G2, G3:

- $*$ is associative.
- e is a neutral element such that $\forall a \in G: a * e = e * a = a$ (minimal version: $a * e = a$).
- Every $a \in G$ has an inverse $\hat{a}$ such that $a * \hat{a} = \hat{a} * a = e$ (minimal version: $a * \hat{a} = e$).

Examples being $(\mathbb{Z}/\mathbb{Z}_m/\mathbb{Q}/\mathbb{R}; +, -, 0)$, $((\mathbb{Q}/\mathbb{R}) \setminus \{0\}; \cdot, ^{-1}, 1)$. A group is called abelian (commutative) if $\forall a, b \in G: a * b = b * a$

Lemma (5.3) $\forall a, b, c \in G, \langle G; *, \hat{\cdot}, e \rangle$:

- $(\hat{\hat{a}}) = a$
- $\hat{a * b} = \hat{b} * \hat{a}$
- $a * b = a * c \implies b = c$ (left) and $b * a = c * a \implies b = c$ (right)
- $a * x = b$ and $x * a = b$ have a unique solution x for any a, b .

Subgroups $\langle H; *, \hat{\cdot}, e \rangle$ is a subset of a group $\langle G; *, \hat{\cdot}, e \rangle$ if:

- $\forall a, b \in H. a * b \in H$
- $e \in H$
- $\forall a \in H. \hat{a} \in H$

Every group has itself and $\{e\}$ as trivial subgroups.

Number of subgroups: $\sum_{a \mid n \wedge b \mid m} \gcd(a, b)$ (non-script)

Direct Product The direct product of n groups $\langle G_i; *_i \rangle, \dots, \langle G_n; *_n \rangle$ is the algebra $\langle G_1 \times \dots \times G_n; * \rangle$. The operation $*$ is applied component-wise, i.e.,

$$(a_1, \dots, a_n) * (b_1, \dots, b_n) = a_1 *_1 b_1, \dots$$

The order of an $(a, b) \in G \times H$ is $\text{lcm}(\text{ord}(a), \text{ord}(b))$. If the two orders are relatively prime, the order of (a, b) is the product of their orders. If G, H cyclic with relatively prime orders, $G \times H$ cyclic.

$$|G \times H| = |G| \cdot |H|$$

Homo- and Isomorphisms Given $\langle G; *, \hat{\cdot}, e \rangle, \langle H; *, \sim, e' \rangle$, a function $\varphi: G \rightarrow H$ is a homomorphism if:

$$\varphi(a * b) = \varphi(a) * \varphi(b)$$

A homomorphism must satisfy $\varphi(e) = e', \varphi(\hat{a}) = \widehat{\varphi(a)}$ – both **neutral elements and inverses are preserved**.

Example – log is a homomorphism from $(\mathbb{R}^{>0}; \cdot) \rightarrow \langle \mathbb{R}; + \rangle$ since $\log(a \cdot b) = \log(a) + \log(b)$

Theorem (5.7) A cyclic group of order n is isomorphic to $\langle \mathbb{Z}_n; \oplus \rangle$ (and hence abelian).

$$\mathbb{Z}_{nm} \cong \mathbb{Z}_n \times \mathbb{Z}_m \text{ if } \gcd(n, m) = 1$$

Euler's Totient and $\mathbb{Z}_m^*$ $= \{a \in \mathbb{Z}_m : \gcd(a, m) = 1\}$; $\varphi(p^e) = (p-1)p^{e-1}$ for primes p .

$$\varphi(m) = |\mathbb{Z}_m^*| = \prod_{i=1}^r (p_i - 1)p_i^{e_i-1} = m \prod_{p \mid m, \text{prime}(p)} \left(1 - \frac{1}{p}\right)$$

$\langle \mathbb{Z}_m; \cdot, ^{-1}, 1 \rangle$ is a group.

Explain why.

$a^{\varphi(m)} \equiv 1 \pmod{m}$ and hence $a^{p-1} \equiv 1 \pmod{p}$ for primes p (i.e., where $\gcd(a, m) = 1$). Also, $a^x \pmod{m} \equiv a^{x \bmod \varphi(m)} \pmod{m}$ if $\gcd(a, n) = 1$.

4.1 Order and Cyclic Groups

Order

- On Elements: The m such that $a^m = e$. $\text{ord}(a) = \infty$ if there is no such m . If $\text{ord}(a) = 2$, $a^{-1} = a$.
- On Groups: $|G|$ (and for every element, we have $a^{|G|} = e$)

Cyclic Groups

- $a^m = a^{R_{\text{ord}(a)}(m)}$
- $\langle a \rangle = \{a^n : n \in \mathbb{Z}\}$ is the group generated by an $a \in G$.

A group generated by an element $g \in G$ is called cyclic. Cyclic groups of order n are **commutative** and isomorphic to $\langle \mathbb{Z}_n; + \rangle$.

The order of a subgroup divides the group order. Thus, for a finite G , $\forall g \in G$, $\text{ord}(g) \mid \text{ord}(G)$, and $a^{|G|} = e$.

The number of generators is $\varphi(|G|)$ (non-script) and the set of generators is the coprime powers of an existing generator g : $\{g^i : \gcd(i, |G|) = 1\}$

Every group of prime order is cyclic, and every element except for e is a generator. $\langle \mathbb{Z}_p; + \rangle$ is always cyclic. Groups $\mathbb{Z}_a \times \mathbb{Z}_b$, are cyclic if $\gcd(a, b) = 1$.

4.2 RSA

Define a group G and find two large primes p, q .

- $n = pq$
- $|G| = |\mathbb{Z}_n^*| = |\mathbb{Z}_{pq}^*| = \varphi(n) = (p-1)(q-1)$

Then, let $e \in \mathbb{Z}$ be relatively prime to $|G|$ and $d \equiv e^{-1} \pmod{|G|}$. Then, $x \mapsto x^e$ is a bijection and for all ciphertext messages $c = x^e$, $x = c^d = x^{ed}$.

Proof: $ed = k|G| + 1$, so $x^{ed} = x^{k|G|+1} = (x^{|G|})^k \cdot x = x$.

Application

- Select e and find $d \equiv e^{-1} \pmod{|G|}$
- Publish your (n, e)
- Other party calculates $c = R_n(m^e)$ and transmits it
- You calculate $m = R_n(c^d)$

4.3 Rings, Fields

4.3.1 Rings

A ring $\langle R; +, -, 0, \cdot, 1 \rangle$ is an algebra for which:

- $\langle R; +, -, 0 \rangle$ is a commutative group
- $\langle R; \cdot, 1 \rangle$ is a monoid
- $a(b+c) = (ab) + (ac)$ and $(b+c)a = (ba) + (ca)$ (left, right distributivity)

A ring is called commutative if multiplication is commutative.

Additionally:

- $0a = a0 = 0$
- $(-a)b = -(ab)$
- $(-a)(-b) = ab$
- If $|R| > 1$, we have $1 \neq 0$

A finite field of order q exists $\iff q$ is a prime power p^k . $\text{GF}^*(q)$ has an order of $q-1$ with $\varphi(q-1)$ generators.

In a field of order p^k , the characteristic (number of times the mult. id needs to be added to get add. id) of the field is the prime base p , so $(x+y)^p = x^p + y^p$ holds (requires proof).

All fields of order q are isomorphic.

A field cannot contain two different finite subfields with the same order (requires proof).

Unit u is a unit if it is invertible such that $uv = vu = 1$. The units of R are denoted R^* , which is itself a multiplicative group. 0 is never invertible in a mult. group.

For an integral domain D , the number of units in $D^* = D[x]^*$.

Divisibility $a \mid b \iff b = ac, c \in R$. In commutative rings, we have:

- $a \mid b \wedge b \mid c \implies a \mid c$
- $a \mid b \implies a \mid bc$
- $a \mid b \wedge a \mid c \implies a \mid (b+c)$

Alternative definition of $\gcd(a, b) = d$: $d \mid a \wedge d \mid b \wedge \forall c(c \mid a \wedge c \mid b) \rightarrow c \mid d$

Integral Domain An element $a \neq 0$ is a zero divisor if $ab = 0$ for some $b \neq 0$. An integral domain is a nontrivial commutative ring without zero divisors.

Polynomials Rings Denoted $\sum_{i=0}^d a_i x^i$. $\deg(a(x))$ is the greatest i for which $a_i \neq 0$.

$\deg(0) = -\infty$.

- For a commutative ring R , $R[x]$ is also commutative.
- If D is an integral domain, so is $D[x]$
 - The degree of the product of two polynomials is the sum of their degrees
 - The units of $D[x]$ are the constant polynomials that are units of D .

$$\mathbb{R}[x]_{\leq 2+1} \cong \mathbb{C}$$

4.3.2 Fields

A field is a commutative ring F in which every nonzero element is a unit.

- F field $\iff \langle F \setminus \{0\}; \cdot, ^{-1}, 1 \rangle$ abelian group
- $\mathbb{Z}_p$ field $\iff \text{prime}(p)$ – denoted $\text{GF}(p)$

Fields are integral domains. Multiplicative fields exclude 0.

4.3.3 Finite Fields

Example: $F[x]_{m(x)}$, i.e., if $m(x)$ is a polynomial of degree d , then $F[x]_{m(x)} = \{a(x) \in F[x] : \deg(a(x)) < d\}$. Congruence modulo $m(x)$ is an equivalence relation on $F[x]$, and each equivalence class has a unique representative of degree less than $\deg(m(x))$ (script).

If $F[x]$ and $\deg(m(x)) = d$, then $|F[x]_{m(x)}| = q^d$.

$F[x]_{m(x)}$ is a ring under addition and multiplication mod $m(x)$.

Analogous to regular inverses, $a(x)b(x) \equiv 1 \pmod{m(x)}$ if they are inverses of each other and iff $\gcd(a(x), m(x)) = 1$.

$F[x]_{m(x)}$ is a field iff $m(x)$ is irreducible.

4.4 Polynomials

An $a(x) \in F[x]$ is monic/normed if the leading coefficient is 1; irreducible if it is only divisible by constant polynomials and constant multiples of $a(x)$.

The monic $g(x)$ of largest degree such that $g(x) \mid a(x) \wedge g(x) \mid b(x)$ is the $\gcd(a(x), b(x))$.

Division with remainder: $a(x) = b(x)q(x) + r(x)$

Evaluation If $c(x) = a(x) + b(x)$, then $c(\alpha) = a(\alpha) + b(\alpha)$ for any α . Same for multiplication.

Roots α root $\iff (x-\alpha) \mid a(x)$

Polynomials of deg 2 or 3 are irreducible $\iff$ they have no roots.

A nonzero polynomial $a(x)$ of degree d has at most d roots and is determined by any unique $d+1$ values of $a(x)$.

$$F[x]_{m(x)} \text{ Example: } 5x^3 - 2x + 1 \equiv_{3x^2+2} 8x^3 + 1 \equiv_{3x^2+2} -\frac{16}{3}x + 1$$

This structure is a ring under addition and multiplication mod $m(x)$.

4.4.1 Lagrange Interpolation

$$a(x) = \sum_{i=0}^n y_i L_i(x)$$

where $L_i(x) = \prod_{j=0, j \neq i}^n \frac{x-x_j}{x_i-x_j}$.

e.g., given $(0, 1), (1, 3), (2, 2)$, we have:

$$\begin{aligned}L_0(x) &= \frac{(x-1)(x-2)}{(0-1)(0-2)}, L_1(x) = \frac{(x-0)(x-2)}{(1-0)(1-2)}, L_2(x) = \frac{(x-0)(x-1)}{(2-0)(2-1)} \\ \implies a(x) &= 1 \cdot L_0(x) + 3 \cdot L_1(x) + 2 \cdot L_2(x)\end{aligned}$$

4.5 ECC

An (n, k) -encoding function E over the alphabet $\mathcal{A}$ is an injective function mapping a list $(a_0, \dots, a_{k-1}) \in \mathcal{A}^k$ of k symbols to a list $(c_0, \dots, c_{n-1}) \in \mathcal{A}^n$ of $n > k$ encoded symbols in $\mathcal{A}$ (a codeword).

Formally,

$$E: \mathcal{A}^k \rightarrow \mathcal{A}^n : (a_0, \dots, a_{k-1}) \mapsto E((a_0, \dots, a_{k-1})) = (c_0, \dots, c_{n-1})$$

Similarly, a decoding function $D: \mathcal{A}^n \rightarrow \mathcal{A}^k$.

The set of codewords $\mathcal{C} = \mathcal{I}(E)$.

An (n, k) error correcting code over an alphabet $\mathcal{A}$ with $|\mathcal{A}| = q$ is a subset of $\mathcal{A}^n$ of cardinality q^k .

Hamming Distance The number of positions at which two strings differ.

Minimum Distance Minimum Hamming Distance.

Error Correction Tolerance A decoding function is t -error-correcting if for arbitrary $(a_0, \dots, a_{k-1})$, the original message can be recovered. A **code** is t -error-correcting if there are E, D with $C = \mathcal{I}(E)$ where D is t -error-correcting.

A code with minimum distance d is t -error-correcting iff $d \geq 2t + 1$.

Polynomial Codes If the code is over an $\mathcal{A} = \text{GF}(q)$, one can use:

$$E((a_0, \dots, a_{k-1})) = (a(a_0), \dots, a(a_{n-1}))$$

where $a(x) = a_{k-1}x^{k-1} + \dots + a_1x + a_0$, giving a $d_{\min}$ of $n - k + 1$.

5 Logic

5.1 Proof Systems

A proof system $\pi = (S, P, \tau, \varphi)$ is:

- Sound if no incorrect $(\tau(s) = 0)$ statements have a $p \in P$ with $(\varphi(s, p) = 1)$
- Complete if for every $s \in S$ with $(\tau(s) = 1)$, there is a $p \in P$ with $(\varphi(s, p) = 1)$

Soundness can be stated as $F \vdash G \implies F \models G$. Def. 6.21: A derivation rule is correct if for every set M of formulae and every formula $F, M \vdash F \implies M \models F$ and a calculus is sound if this holds for all rules (Def. 6.22).

Completeness can be stated as $F \models G \implies F \vdash G$. Def. 6.22: A calculus is complete if this holds $\forall M, F$.

τ is the truth function defined as $\tau : \mathcal{S} \rightarrow \{0, 1\}$, φ is the validation function $\mathcal{S} \times \mathcal{P} \rightarrow \{0, 1\}$.

Syntax Alphabet A of allowed symbols and specification of which strings in A^* are formulae. Def. 6.31:

Syntax of Propositional logic:

- An atomic formula $A_i, i \in \mathbb{N}$ is a formula.
- If F, G are formulae, then $\neg F, (F \wedge G), (F \vee G)$ are.

Syntax of Predicate logic:

- A variable symbol is an $x_i, i \in \mathbb{N}$
- A function symbol is an f_i^k with $i, k \in \mathbb{N}$ and k being the number of arguments. $k = 0$ is a constant.
- A predicate symbol is a P_i^k and analogous to a f .
- A formula is defined inductively:
 - $\forall i, k$, if $t_1, \dots, t_k$ are terms, then $P_i^k(t_1, \dots)$ is an atomic formula
 - If F, G are formulae, then $\neg F, (F \wedge G), (F \vee G)$ are
 - If F is a formula, then for any $i, \forall x_i F$ and $\exists x_i F$ are

Interpretation

- Consists of a set $\mathcal{Z} \subseteq A$ of symbols of A , a domain for each symbol in $\mathcal{Z}$, and a function that assigns each symbol in $\mathcal{Z}$ a value in its associated domain.
- Suitable if it assigns a value to all free occurrences in F
- (Def. 6.36) Written $\mathcal{A} = (U, \phi, \psi, \xi)$
 - $\mathcal{A}(t)$ of a term t is defined recursively as:
 - t variable, then $\mathcal{A}(t) = \xi(x_i)$
 - t function $f(t_1, \dots, t_k)$ for terms t_i and k -ary function f , then $\mathcal{A}(t) = \phi(f)(\mathcal{A}(t_1), \dots)$
 - $\mathcal{A}(F)$ of a formula is defined recursively as:
 - F in form $F = P(t_1, \dots, t_k)$ for terms t_i and k -ary P , then $\mathcal{A}(F) = \psi(P)(\mathcal{A}(t_1), \dots)$
 - If F is of the form $\forall xG$ or $\exists xG$, then let $\mathcal{A}_{[x \rightarrow u]}$ for $u \in U$ be the same structure as $\mathcal{A}$, except that $\xi(x)$ is overwritten by u :
 - $\mathcal{A}(\forall xG) = \begin{cases} 1 & \text{if } \mathcal{A}_{[x \rightarrow u]}(G) = 1 \text{ for all } u \in U \\ 0 & \text{else} \end{cases}$
 - Analog for $\exists$ and some $u \in U$.

Model A suitable $\mathcal{A}$ for which F is true is a model, and is written $\mathcal{A} \models F$. If A is not a model, one writes $\mathcal{A} \not\models F$.

Universal Instantiation (L6.11) For any formula and term, we have $\forall xF \models F[x/t]$, allowing the quantifier to be eliminated.

Satisfiability A formula F is satisfiable if there exists a model for F , otherwise unsatisfiable.

5.2 Semantics

The definition of the meaning of a formula. Defines under which conditions a formula is true. Defines a function *free* which assigns each formula $F = (f_1, \dots, f_k) \in A^*$ a subset $free(F) \subseteq \{1, \dots, k\}$ of the indices. If $i \in free(F)$, f_i occurs free in F .

- Defines a function σ assigning to each formula F and suitable interpretation $\mathcal{A}$ a truth value $\mathcal{A}(F) \in \{0, 1\}$.

5.3 Semantics of $\wedge, \vee, \neg$

If F, G are formulae, then $\neg F, (F \wedge G), (F \vee G)$ also are.

Lemma 6.1 Rules from Lemma 2.1, but for formulae (F, G) and with the following additions

- $F \vee \top \equiv \top$ and $F \wedge \top \equiv F$
- $F \vee \perp \equiv F$ and $F \wedge \perp \equiv \perp$
- $F \vee \neg F \equiv \top$ and $F \wedge \neg F \equiv \perp$

Lemma 6.3 The following statements are equivalent:

- $\{F_1, F_2, \dots, F_k\} \models G$
- $(F_1 \wedge F_2 \wedge \dots \wedge F_k) \rightarrow G$ is a tautology
- $\{F_1, F_2, \dots, F_k, \neg G\}$ is unsatisfiable

5.4 Resolution Calculus

Given a set of clauses in CNF, $\{A, B\} \vdash_{res} A \setminus \{C\} \cup B \setminus \{C\}$. e.g., $\{\{A, B, C\}, \{C\}\} \vdash_{res} \{A, B\}$

A set of formulas M is unsatisfiable $\iff \mathcal{K}(M) \vdash \emptyset$ (i.e., derive the empty set to show unsatisfiability). To show a tautology (i.e., that a formula is valid), derive the empty set from

$\neg F$. To show that a formula F can be derived from a set of clauses M , show that $\{M, \neg F\} \vdash_{res} \emptyset$

5.5 Normal Forms

- **Rectified Form** if no variable is both bounded and free. So $(\exists xP(x)) \vee Q(x) \implies (\exists uP(u)) \vee Q(x)$. Rename **bound** variables only.
- **Prenex Normal Form** if all quantifiers are in front (rename variables bound to quantifiers to prevent collisions and then rearrange)
- **CNF** an AND of ORs $(C_1 \vee C_2) \wedge (C_3 \vee C_4)$
- **DNF** an OR of ANDs $(C_1 \wedge C_2) \vee (C_3 \wedge C_4)$

5.6 Theorem 6.12

$$\neg \exists x \forall y P(y, x) \leftrightarrow \neg P(y, y) [\equiv \forall x \exists y P(y, x) \leftrightarrow P(y, y)]$$

Example: There exists no set that contains all sets S that do not contain themselves, i.e., $\{S : S \notin S\}$ is not a set.

6 Appendices

6.1 Tables

Primes 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181

Modular Inverses cell = $a^{-1} \bmod m$, where rows are m , cols are a .

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	1		1		1		1		1		1		1		1		1		1		1		1		1
3	1	2		1	2		1	2		1	2		1	2		1	2		1	2		1	2		1
4	1		3		1		3		1		3		1		3		1		3		1		3		1
5	1	3	2	4		1	3	2	4		1	3	2	4		1	3	2	4		1	3	2	4	
6	1				5		1				5		1			5		1				5		1	
7	1	4	5	2	3	6		1	4	5	2	3	6		1	4	5	2	3	6		1	4	5	2
8	1		3		5		7		1		3		5		7		1		3		5		7		1
9	1	5		7	2		4	8		1	5		7	2		4	8		1	5		7	2		4
10	1		7				3		9		1		7			3		9		1		7			
11	1	6	4	3	9	2	8	7	5	10		1	6	4	3	9	2	8	7	5	10		1	6	4
12	1				5		7				11		1			5		7				11		1	
13	1	7	9	10	8	11	2	5	3	4	6	12		1	7	9	10	8	11	2	5	3	4	6	12
14	1		5		3				11		9		13		1		5		3				11		9
15	1	8		4			13	2			11		7	14		1	8		4			13	2		
16	1		11		13		7		9		3		5		15		1		11		13		7		9
17	1	9	6	13	7	3	5	15	2	12	14	10	4	11	8	16		1	9	6	13	7	3	5	15
18	1				11		13				5		7			17		1				11			13
19	1	10	13	5	4	16	11	12	17	2	7	8	3	15	14	6	9	18		1	10	13	5	4	16
20	1		7				3		9		11		17			13		19		1		7			
21	1	11		16	17			8		19	2		13			4	5		10	20		1	11		16
22	1		15		9		19		5				17		3		13		7		21		1		15
23	1	12	8	6	14	4	10	3	18	7	21	2	16	5	20	13	19	9	17	15	11	22		1	12
24	1				5		7				11		13				17		19				23		1

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
25	1	13	17	19		21	18	22	14		16	23	2	9		11	3	7	4		6	8	12	24	

Euler's Totient Table cell = $\varphi(\text{row} + \text{col})$

+	1	2	3	4	5	6	7	8	9	10
0	1	1	2	2	4	2	6	4	6	4
10	10	4	12	6	8	8	16	6	18	8
20	12	10	22	8	20	12	18	12	28	8
30	30	16	20	16	24	12	36	18	24	16
40	40	12	42	20	24	22	46	16	42	20
50	32	24	52	18	40	24	36	28	58	16
60	60	30	36	32	48	20	66	32	44	24
70	70	24	72	36	40	36	60	24	78	32
80	54	40	82	24	64	42	56	40	88	24
90	72	44	60	46	72	32	96	42	60	40

		Monoid	Group	Abelian Group	Ring	Commutative Ring	Integral Domain	Field
additive	Closure	●	●	●	●	●	●	●
	Associative	●	●	●	●	●	●	●
	Identity	●	●	●	●	●	●	●
	Inverse		●	●	●	●	●	●
	Commutative			●	●	●	●	●
multiplicative	Closure				●	●	●	●
	Associative				●	●	●	●
	Distributive				●	●	●	●
	Commutative					●	●	●
	Identity				●	●	●	●
	No zero-divisors						●	●
	Inverse							●

6.2 Irreducible Polynomials

GF(2)

x^4	x^3	x^2	x	c
			1	0
			1	1
		1	1	1
	1	0	1	1
	1	1	0	1
1	0	0	1	1
1	1	0	0	1
1	1	1	1	1

GF(3)

x^3	x^2	x	c
		1	0,1,2
	1	0	1
	1	1	2
	1	2	2
1	0	2	1
1	0	2	2
1	1	0	2
1	1	1	2
1	1	2	1
1	2	0	1
1	2	1	1
1	2	2	2
...			

GF(5)

x^2	x	c
	1	0,1,2,3,4
1	0	2,3
1	1	1,2
1	2	3,4
1	3	3,4
1	4	1,2
...		

GF(7)

x^2	x	c
	1	0,1,2,3,4,5,6
1	0	1,2,4
1	1	3,4,6
1	2	2,3,5
...		

6.3 Group Cardinalities

1	$\mathbb{Z}_1$	2	$\mathbb{Z}_2$	3	$\mathbb{Z}_3$
4	$\mathbb{Z}_4, \mathbb{Z}_2 \times \mathbb{Z}_2$	5	$\mathbb{Z}_5$	6	$\mathbb{Z}_6$
7	$\mathbb{Z}_7$	8	$\mathbb{Z}_8, \mathbb{Z}_2 \times \mathbb{Z}_4, \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$	9	$\mathbb{Z}_9, \mathbb{Z}_3 \times \mathbb{Z}_3$
10	$\mathbb{Z}_{10}$				

6.4 Groups and Generators

2	1	3	1, 2
4	1, 3	5	1, 2, 3, 4
6	1, 5	7	1, 2, 3, 4, 5, 6
8	1, 3, 5, 7	9	1, 2, 4, 5, 7, 8
10	1, 3, 7, 9	11	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
12	1, 5, 7, 11	13	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
14	1, 3, 5, 9, 11, 13	15	1, 2, 4, 7, 8, 11, 13, 14
16	1, 3, 5, 7, 9, 11, 13, 15	17	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16

$\mathbb{Z}_n$ has $\varphi(n)$ generators. An isomorphism is defined uniquely by which generator is mapped to *one* other.

7 Examples

7.1 Sets

Are we dealing with infinite sets?

Does every element have a mapping? Do we need a default case in the function definition?

7.1.1 $(A; \rho)$ poset. Prove $(A; \hat{\rho})$ poset

Working backwards from inverse:

- R: $a\hat{\rho}a \implies a\rho a$
- A: $a\hat{\rho}b \wedge b\hat{\rho}a \implies b\rho a \wedge a\rho b \implies a = b$
- T: $a\hat{\rho}b \wedge b\hat{\rho}c \implies b\rho a \wedge c\rho b \implies c\rho b \wedge b\rho a \implies c\rho a \implies a\hat{\rho}c$

7.1.2 Uncountability

- Find injection $f : \{0, 1\}^\infty \rightarrow A$
- Show f is **totally and well-defined**
- Prove injectivity

Very useful: Prime factorization – e.g., for listing the finite number of indices at which an $f(x) = 1$, we can define $g(f) = 2^{x_1} 3^{x_2} \dots$ since there are infinitely many primes.

TODO: FIX

- Complement: Find uncountable $B, A \subseteq B$. Show $B \setminus A$ countable. Then:

$$\underbrace{A \cup (B \setminus A)}_{\leq \aleph} = \underbrace{B}_{\leq \aleph} \quad \text{uncountable}$$

7.1.3 Prove that $A \subseteq B \iff P(A) \subseteq P(B)$

$\implies$ Direction: Consider an arbitrary $a \in P(A)$.

$\implies a \subseteq A$ [Def. Power Set] $\implies a \subseteq B$ [Transitivity $\subseteq$, Assumption $a \subseteq B$] $\implies a \in P(B)$

$\Leftarrow$ Direction:

$A \subseteq A \implies A \in P(A)$ [Def. Power Set] $\implies A \in P(B)$ [Assumption] $\implies A \subseteq B$

7.1.4 Let A be a nonempty set and let $f : A \rightarrow \mathcal{P}(A)$ be a function. Show that $Z_f := \{x \in A : x \notin f(x)\}$ is not in the image of f .

For all $x \in A$, we either have $x \in f(x) \vee x \notin f(x)$. Case distinction:

- $x \in f(x)$ – that implies $f(x) \neq Z_f$ [since $x \in A \wedge x \in f(x)$]
- $x \notin f(x)$ – that implies $x \in Z_f \implies f(x) \neq Z_f$ [Def. Set Equality] $\implies Z_f \notin \text{Im}(f)$

Which can be used to prove $A \prec \mathcal{P}(A)$:

7.1.4.1 Simple

Since $Z_f \in \mathcal{P}(A)$, but $Z_f \notin \text{Im}(f)$, there is no surjection onto $\mathcal{P}(A)$ and hence $A \nsim \mathcal{P}(A)$. But there is an injection $g : A \rightarrow \mathcal{P}(A)$ with $g(x) = \{x\}$, so $A \preceq \mathcal{P}(A)$. Combining the two, $A \prec \mathcal{P}(A)$.

7.1.5 Set of all reflexive relations on $\mathbb{N} \succ \mathbb{N}$

We give an injection from $\{0, 1\}^\infty$ to the set of reflexive relations on $\mathbb{N}$ to show that this set is uncountable. Let $s \in \{0, 1\}^\infty$ be arbitrary. We then define $f(s)$ as $R = \{(a, a + 1) \mid s_a = 1\} \cup \{(x, x) \mid x \in \mathbb{N}\}$ where s_i is the i -th bit of s .

We now show that f is injective. Let $s, t \in \{0, 1\}^\infty$ with $s \neq t$. Let $i \in \mathbb{N}$ be the smallest i such that $s_i \neq t_i$. Without loss of generality assume $s_i = 1$ and $t_i = 0$ (the opposite case can be shown analogously). It follows that $(i, i + 1) \in f(s)$ but $(i, i + 1) \notin f(t)$. Thus $f(s) \neq f(t)$. Hence f is injective and the set of all reflexive relations on $\mathbb{N}$ must be uncountable since it dominates $\{0, 1\}^\infty$ which itself is uncountable.

7.1.6 IJ/SJ of Composed Functions

$f : A \rightarrow B$ and $g : B \rightarrow C$. $(g \circ f) : A \rightarrow C$.

7.1.6.1 1. Injectivity

Theorem: If $g \circ f$ is injective, then f is Injective. Proof:

Goal: Show that $f(x_1) = f(x_2) \implies x_1 = x_2$. Assume $f(x_1) = f(x_2)$ for some $x_1, x_2 \in A$.

Apply g to both sides: $g(f(x_1)) = g(f(x_2))$. By definition of composition: $(g \circ f)(x_1) = (g \circ f)(x_2)$. Since $g \circ f$ is injective (given), we know $x_1 = x_2$. So f injective.

7.1.6.2 2. Surjectivity

Theorem: If $g \circ f$ is surjective, then g is surjective. Proof:

Goal: Show that for any $z \in C$, there exists a $y \in B$ such that $g(y) = z$. Pick an arbitrary $z \in C$. Since $g \circ f$ is surjective (given), there exists an $x \in A$ such that $(g \circ f)(x) = z$.

Rewrite this as: $g(f(x)) = z$. Let $y = f(x)$. Note that $y \in B$ (since the codomain of f is B). Substitute y : we have found a $y \in B$ such that $g(y) = z$. So g surjective.

7.2 Algebra

$0a = 0$:

$$\begin{aligned} a0 &= a(0 + 0) \text{ [0 id]} \\ &= a0 + a0 \text{ [L. Dist]} \\ a0 - a0 &= a0 + a0 - a0 \text{ [Subtract]} \\ &= a0 \end{aligned}$$

$(-a)b = -(ab)$: Due to the uniqueness of inverses, it suffices to prove that ab (the inverse of $- (ab)$) is also an inverse of $(-a)b$. Verify: $(-a)b + ab = (-a + a)b = 0b = 0$.

Commutativity of addition from axioms Assume not commutative. But $(1 + 1)(x + y) = (x + y) + (x + y) = x(1 + 1) + y(1 + 1)$, so must be commutative.

$(-1)(-1) = 1$

$$\begin{aligned} 1 - 1 &= 0 \implies (1 - 1)(-1) = 0(-1) \implies (1 - 1)(-1) = 0 \\ &\implies -1 + (-1)(-1) = 0 \implies (-1)(-1) = 1 \end{aligned}$$

7.2.1 Find a generator of $\text{GF}(3)[x]_{x^2+2x+1}^*$

Since $|G| = 6$, $\text{ord}(a) \in \{1, 2, 3, 6\}$ for any $a \in G$. By trial and error, the order of x is 6 because $x = x, x^2 = x + 2, x^3 = 2$, so it must be 6 and thus a generator.

7.2.2 Non-Trivial Homomorphism $f : (\mathbb{Z}_9; +) \rightarrow (\mathbb{Z}_{12}; +)$

- The map between the cyclic groups must divide both $|\mathbb{Z}_9|, |\mathbb{Z}_{12}|$. So use $3 = \text{gcd}(9, 12)$.
- So subgroup G of $\mathbb{Z}_{12}$ with $\text{ord}(G) | 3$
- So element $\in \mathbb{Z}_{12}$ with $\text{ord} = 3$.
- The element k generating a subgroup of order d in $\mathbb{Z}_m$ is given by $k = \frac{m}{d}$, so $k = 4$
- So map generator $1 \in \mathbb{Z}_9 \rightarrow 4 \in \mathbb{Z}_{12}$, giving $0 \rightarrow 0, 1 \rightarrow 4, 2 \rightarrow 8$ and $f(x) = 4x \pmod{12}$.

7.2.3 Non-constant Polynomial $a(x) \in F[x]$ such that F has no roots

- Define $c(x) = \prod_{b \in F} (x - b)$. $\forall b \in F, c(b) = 0$.
- Define $a(x) = c(x) + 1$. This has no roots in $F[x]$.

7.2.4 Fermat’s Little Theorem

Useful for CRT; same as with group orders, given a prime p and integer a not divisible by p , we have:

$$a^{p-1} \equiv 1 \pmod{p}$$

Example – Calculating $3^{234} \pmod{7}$. $3^6 \equiv 1 \pmod{7}$, so $3^{234} = (3^6)^{39} \equiv 1^{39} \equiv 1$.

7.2.5 Cyclic Groups are Commutative

Any element can be represented as a power of a generator: $g^a \cdot g^b = g^{a+b} = g^{b+a} = g^b \cdot g^a$

7.2.6 Decompose $a(y) = xy^3 + xy^2 + (x + 1)y + x \in \text{GF}(2)[x]_{x^2+x+1}[y]$ into irreducible factors

In $\text{GF}(2)$, $y \in \{0, 1, x, x + 1\}$. So substitute each possibility into y to find a root.

7.2.7 Isomorphism between $\langle \mathbb{Z}_3; \oplus \rangle \times \langle \mathbb{Z}_{35}; \oplus \rangle \rightarrow \langle \mathbb{Z}_{21}; \oplus \rangle \times \langle \mathbb{Z}_5; \oplus \rangle$

$\text{gcd}(3, 35) = 1$, so $f : \langle \mathbb{Z}_{105}; \oplus \rangle \rightarrow \langle \mathbb{Z}_3; \oplus \rangle \times \langle \mathbb{Z}_{35}; \oplus \rangle, x \mapsto (R_3(x), R_{35}(x))$ is an isomorphism.

$\text{gcd}(21, 5) = 1$, so $g : \langle \mathbb{Z}_{105}, \oplus \rangle \rightarrow \langle \mathbb{Z}_5; \oplus \rangle \times \langle \mathbb{Z}_{21}; \oplus \rangle, x \mapsto (R_5(x), R_{21}(x))$ is an isomorphism.

Thus, $g \circ f^{-1}$ is such an isomorphism.

Proof (!!!): As stated in the script, the CRT implies a one-to-one correspondence (isomorphism) between $\mathbb{Z}_m$ and its direct product components provided they are pairwise relatively prime.

7.2.8 Number of Isomorphisms between two cyclic groups of $\text{ord} > 0$

The number of isomorphisms between two cyclic groups of order n is $\varphi(n)$.

Proof Sketch:

- $\psi : G \rightarrow H$ is uniquely determined by the image $\psi(a)$ of a generator. ($g \in G = a^k$, apply definition HM and powers)
- An isomorphism must map a generator to a generator. For ψ to be a bijection, it must be surjective, so its image $\text{Im}(\psi)$ must equal H . Since $\text{Im}(\psi) = \{\psi(a)^k \mid k \in \mathbb{Z}\}$, the set of powers of $\psi(a)$ must contain all n elements of H . This occurs if and only if the element $\psi(a)$ has order n , which by definition makes $\psi(a)$ a generator of H .
- Every cyclic group is IM to $(\mathbb{Z}_n; +)$. Generators of such are $\text{gcd}(a, n) = 1$, the number of which is defined as the Euler function. So exactly $\varphi(n)$ generators/elements that can serve as image $\varphi(a)$.
- Prove well-defined, HM, bijectivity

7.2.9 Is $\mathbb{Z}_n^*$ cyclic?

$\mathbb{Z}_n^*$ cyclic $\iff n \in \{1, 2, 4, p^k, 2p^k\}$ for some prime k .

7.2.10 Zero divisors in ring $\langle \mathbb{Z}_n; \oplus, \otimes \rangle$

$|\mathbb{Z}_n| - \varphi(n) - 1$ (to remove 0)

7.2.11 Claim to Disprove: For any groups G and H , all subgroups of $G \times H$ are of the form $G' \times H'$ (where $G' \leq G$ and $H' \leq H$).

7.2.11.1 Disproof Strategy

We construct a counterexample using a “diagonal” subgroup, which links the choices of elements from G and H , preventing them from being formed by independent factors.

7.2.11.2 Counterexample

Let $G = \mathbb{Z}_2 = \{0, 1\}$ and $H = \mathbb{Z}_2 = \{0, 1\}$. The direct product is:

$$G \times H = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$$

Consider the **diagonal subgroup** D :

$$D = \{(0, 0), (1, 1)\}$$

- Verify D is a subgroup:**
 - Contains identity $(0, 0)$.
 - Closed under addition: $(1, 1) + (1, 1) = (0, 0)$.
 - Contains inverses (elements are self-inverse).
- Show D is not a product $G' \times H'$:**
 - If $D = G' \times H'$, then since $(1, 1) \in D$, we must have $1 \in G'$ and $1 \in H'$.
 - This implies $G' = \{0, 1\}$ and $H' = \{0, 1\}$.
 - The product $G' \times H'$ would be the full group $\{(0, 0), (0, 1), (1, 0), (1, 1)\}$.
 - However, D only has 2 elements.

Conclusion: Since $\{(0, 0), (1, 1)\} \neq G \times H$, the claim is false.

7.2.12 Let F be a finite field. Prove there are infinitely many irreducible polynomials in $F^n[x]$

Assume finitely many. List them as $\{p_1(x), \dots, p_n(x)\}$. Construct a new polynomial $f(x) = \prod_{i=1}^n (p_i(x) + 1)$. Since $\deg(f(x)) \geq 1$, at least one irreducible factor. This factor is, by construction, in our list. But dividing this polynomial by any of the listed ones gives a remainder of 1. So not divisible, a contradiction. The irreducible factor must exist, but wasn’t in our list. So list infinite.

7.2.13 $\ker(\varphi) = \{e\} \iff \varphi$ injective

$\implies$: Assume $\ker(\varphi) = \{e_1\}$. We know $\varphi(e_1) = e_2$. Assume not injective, reach contradiction with more elements in $\ker$.

$\Leftarrow$: Assume injective. We know $\varphi(e_1) = e_2$. Use definition of inj. to show other elements in kernel equal to e_1 .

7.2.14 Is $p(y) = y^2 + 5xy + 5 \in \mathbb{Z}_7[x]_{x^2+1}[y]$ reducible? Factors?

Note that by the definition of the field, we have $x^2 + 1 \equiv 0 \implies x^2 \equiv -6$.

Calculate the discriminant: $(5x)^2 - 4(1)(5) = 25x^2 - 20 \equiv 4x^2 - 6$. Substitute $x^2 = 6 \implies 4(6) - 6 > 0$, so root. Solving: $\sqrt{18} \equiv \sqrt{4} = \pm 2$. $y = \frac{-5x \pm 2}{2} \implies y = 4(2x \pm 2) \implies y \in \{x + 1, x + 6\} \implies p(y) = (y - (x + 1))(y - (x + 6))$.

7.3 Logic

7.3.1 Prove $\exists x \forall y P(x, y) \iff \exists x P(x, f(x))$

Let $\mathcal{A}$ be suitable for both LHS and RHS.

$$\begin{aligned} \mathcal{A}(\exists x \forall y \\ P(x, y)) &= 1 \iff \mathcal{A}_{[x \rightarrow u]}(\forall y P(x, y)) = 1 \text{ for some } u \in U^{\mathcal{A}} \text{ [Semantics } \exists] \\ &\iff \mathcal{A}_{[x \rightarrow u][y \rightarrow v]}(P(x, y)) = 1 \text{ for some } u \in U^{\mathcal{A}} \text{ and all } v \in U^{\mathcal{A}} \text{ [Semantics } \forall] \\ &\implies \mathcal{A}_{[x \rightarrow u][y \rightarrow f^{\mathcal{A}}(u)]}(P(x, y)) = 1 \text{ for some } u \in U^{\mathcal{A}} \text{ [} f^{\mathcal{A}}(u) \in U^{\mathcal{A}} \text{]} \\ &\iff \mathcal{A}_{[x \rightarrow u]}(P(x, f(x))) = 1 \text{ for some } u \in U^{\mathcal{A}} \text{ [Semantics } f(x)] \\ &\iff \mathcal{A}(\exists x P(x, f(x))) \text{ [Semantics } \exists] \end{aligned}$$

7.3.2 Prove: $\forall x (F \vee G) \models F \vee (\exists x G)$

7.3.2.1 1. Assumption

Assume the LHS is true in an arbitrary structure $\mathcal{A}$:

$$\mathcal{A}(\forall x (F \vee G)) = 1$$

By the semantics of $\forall$, for every element u in the domain:

$$\mathcal{A}_{[x \rightarrow u]}(F \vee G) = 1$$

By the semantics of $\vee$, this implies that for every u :

$$\mathcal{A}_{[x \rightarrow u]}(F) = 1 \quad \text{or} \quad \mathcal{A}_{[x \rightarrow u]}(G) = 1[*]$$

7.3.2.2 2. Analysis of RHS

We analyze the truth value of $\exists x G$ via two cases.

Case 1: $\mathcal{A}(\exists x G) = 1$
If the existential formula is true, the disjunction $F \vee (\exists x G)$ is trivially true.

$\mathcal{A}(F \vee (\exists x G)) = 1$

Case 2: $\mathcal{A}(\exists x G) = 0$

By definition of $\exists$, this means there is **no** element u such that $\mathcal{A}_{[x \rightarrow u]}(G) = 1$. Therefore, for all u :

$\mathcal{A}_{[x \rightarrow u]}(G) = 0$

Referencing step * above: since G is false for all u , F must be true for all u to satisfy the disjunction.

$\forall u, \mathcal{A}_{[x \rightarrow u]}(F) = 1$

Since this holds for **all** u , it holds for the specific value assigned to x by $\mathcal{A}$ (let's denote the current assignment as $x^{\mathcal{A}}$):

$\mathcal{A}_{[x \rightarrow x^{\mathcal{A}}]}(F) = 1 \implies \mathcal{A}(F) = 1$

Therefore, the RHS is true:

$\mathcal{A}(F \vee (\exists x G)) = 1$

- 7.3.2.3 Conclusion
- In all cases, $\mathcal{A}(\forall x(F \vee G)) = 1 \implies \mathcal{A}(F \vee (\exists x G)) = 1$.
- 7.3.3 Counterexamples in Logic
- Disproving $\forall y \exists x P(x, y) \implies \exists x \forall y P(x, y)$
- Specify $U^{\mathcal{A}} = \mathbb{N}$
 - Specify $P^{\mathcal{A}}(x, y) = 1 \iff x = y$

- 7.3.4 Calculi
- Complete, but not sound: $K_1 = \{ \frac{}{} \}$
 - Sound, but not complete: $K_2 = \emptyset$
- 7.3.5 Prove Resolution Calculus is not complete
- Resolution cannot derive a tautology, e.g., $\{ \} \vdash_{\text{Res}} A \vee \neg A$. While it can prove that something is a tautology by showing $\{ \neg A \vee A \} \vdash \emptyset$, it cannot derive it.

7.3.6 Extend the syntax and semantics of propositional logic by adding $\Diamond$, where $A \Diamond B \iff A \vee B$ is not true

Syntax extension: For two propositional formulae F, G , $F \Diamond G$ is also a formula.

Semantics: $\mathcal{A}(F \Diamond G) = 1 \iff \mathcal{A}(\neg(F \vee G)) = 1$ for suitable $\mathcal{A}$ and formulae F, G .

7.3.7 Prove $\forall x F \models F$

Case distinction, if x not free then trivial. If free:

Since $\mathcal{A}$ suitable for F , it defines $x^{\mathcal{A}}$. So $\mathcal{A}_{[x \rightarrow u]}(F) = 1$ for all $u \in u^{\mathcal{A}} \implies \mathcal{A}_{[x \rightarrow x^{\mathcal{A}}]}(F) = 1$ $[x^{\mathcal{A}} \in U^{\mathcal{A}}] \implies \mathcal{A}(F) = 1$

7.4 Number Theory

Polynomials: $(a - b) | (f(a) - f(b))$

7.4.1 Prove that for all positive integers a, b, c , c a, b relatively prime $\implies \gcd(ab, c) = \gcd(a, c) \gcd(b, c)$

$a = \prod_i p_i^{\alpha_i}, b = \prod_i p_i^{\beta_i}, c = \prod_i p_i^{\gamma_i}, ab = \prod_i p_i^{\alpha_i + \beta_i}$. Thus:

$$\begin{aligned} \gcd(a, c) \gcd(b, c) &= \prod_i p_i^{\min(\alpha_i, \gamma_i)} \prod_i p_i^{\min(\beta_i, \gamma_i)} \quad [\text{Def. gcd}] \\ &= \prod_i p_i^{\min(\alpha_i, \gamma_i) + \min(\beta_i, \gamma_i)} \quad [\text{Exponent Rules}] \\ &= \prod_i p_i^{\min(\alpha_i + \beta_i, \gamma_i)} \quad [*] \\ &= \gcd(ab, c) \quad [\text{Def. gcd}] \end{aligned}$$

Justification for *: We know that a, b are relatively prime, so they do not share any prime factor p_i . Thus, for all $i \in \mathbb{N}$, $\alpha_i = 0$ or $\beta_i = 0$ or $\alpha_i = \beta_i = 0$.

wlog, assume $\alpha_i = 0$. Then, we have:

$$\begin{aligned} \min(\alpha_i, \gamma_i) + \min(\beta_i, \gamma_i) &= \min(0, \gamma_i) + \min(\beta_i, \gamma_i) \\ &= 0 + \min(\beta_i, \gamma_i) \\ &= \min(\beta_i, \gamma_i) \\ &= \min(0 + \beta_i, \gamma_i) \\ &= \min(\alpha_i + \beta_i, \gamma_i) \end{aligned}$$

7.4.2 Prove that lcm distributes over gcd, i.e., $\text{lcm}(a, \gcd(b, c)) = \gcd(\text{lcm}(a, b), \text{lcm}(a, c))$

$$\begin{aligned} \text{lcm}(a, \gcd(b, c)) &= \text{lcm}\left(a, \prod_i p_i^{\min(\beta_i, \gamma_i)}\right) \quad [\text{Def}] \\ &= \prod_i p_i^{\max(\alpha_i, \min(\beta_i, \gamma_i))} \quad [\text{Def}] \\ &= \prod_i p_i^{\min(\max(\alpha_i, \beta_i), \max(\alpha_i, \gamma_i))} \quad [*] \\ &= \gcd\left(\prod_i p_i^{\max(\alpha_i, \beta_i)}, \prod_i p_i^{\max(\alpha_i, \gamma_i)}\right) \quad [\text{Def}] \\ &= \gcd(\text{lcm}(a, b), \text{lcm}(a, c)) \quad [\text{Def}] \end{aligned}$$

- Justification for *:** wlog, assume $y \leq z$ and observe the case distinction:
- $x \leq y \implies \max(x, \min(y, z)) = \max(x, y) = y = \min(y, z) = \min(\max(x, y), \max(x, z))$
 - $y \leq x \leq z \implies \max(x, \min(y, z)) = \max(x, y) = x = \min(x, z) = \min(\max(x, y), \max(x, z))$
 - $z \leq x \implies \max(x, \min(y, z)) = \max(x, y) = x = \min(x, z) = \min(\max(x, y), \max(x, z))$

7.4.3 $a, b, c, d \in \mathbb{N} \setminus \{0\}$. Prove $ab = cd \implies a + b + c + d$ not prime

$$ab = cd \implies \frac{a}{c} = \frac{b}{d}$$

Let $\frac{a}{c}$ be the simplified fraction such that $\gcd(x, y) = 1$. So then $a = ux, c = uy$ and $d = vx, b = vy$. Then substitute into sum equation and factorize to show composite.

7.4.4 $R_{35}(2^{1000})$

CRT with mod 5, mod 7. Per Fermat's Little Theorem, $a^{p-1} \equiv 1 \pmod{p}$.

E.g., $p = 7 \rightarrow$ order 6. $1000 = 166 * 6 + 4$, so $2^{6^{166}}$ full cycles with remainder of 4. Thus $1 * 2^4 \equiv 2$.

7.4.5 RSA: Prove $m^{de} \equiv m \pmod{p} \wedge m^{de} \equiv m \pmod{q} \implies m^{de} \equiv m \pmod{n}$

Define the system $\begin{cases} x \equiv m \pmod{p} \\ x \equiv m \pmod{q} \end{cases}$. p, q coprime with $x_0 = m^{de}, x_1 = m$ being solutions. By the CRT, all solutions are congruent mod $pq = n$. So $m^{de} \equiv m \pmod{n}$

7.4.6 RSA: Prove $\forall m \in \mathbb{Z}_n, m^{de} \equiv m \pmod{p}$

$$\begin{aligned} de - 1 &= k\varphi(n) = k(p-1)(q-1) \implies de = k(p-1)(q-1) + 1 \\ \implies m^{de} &= m^{k(p-1)(q-1)+1} = m^{k(p-1)(q-1)} m = \left(m^{k(q-1)}\right)^{(p-1)} m \equiv ((1)m) \pmod{p} \equiv m \end{aligned}$$

If m is a multiple of p , then $m \equiv 0 \pmod{p} \implies m^{de} \equiv 0 \pmod{p} \equiv m$

7.4.7 Prove $f(p) := x^4 + x^2 + 1$ not prime for all primes p

$$\begin{aligned} f(x) &= x^4 + x^2 + 1 = x^4 + 2x^2 + 1 - x^2 = (x^2 + 1)^2 - x^2 \\ \text{Difference of squares: } a^2 - b^2 &= (a - b)(a + b) \\ f(x) &= (x^2 + 1 - x)(x^2 + 1 + x) \end{aligned}$$

And since both terms are clearly integers greater than 1, their product is a composite number.

8 Edge cases

- Algebras**
- ID, not field: $\langle \mathbb{Z}; 0, +, 1, * \rangle$
 - Ring, not ID: $\langle \mathbb{Z}_4; 0, +, 1, * \rangle$
 - Infinite order
- Sets**
- Finite vs infinite
 - Empty set

9 Exercise Results

9.1 Series 9

9.1.1 Shared Modulus n

Let Alice, Bob have keys $(n, e_A), (n, e_B)$ with $\gcd(e_A, e_B) = 1$. When Charlie encrypts the same message m to both Alice and Bob, sending $c_A \equiv m^{e_A} \pmod{n}, c_B \equiv m^{e_B} \pmod{n}$, a third party can decrypt the ciphertexts.

$c_A \equiv 0 \vee c_B \equiv 0 \pmod{n}$. Then the encrypted message was $m = 0$.

Otherwise, when c_A (wlog, same for c_B) is not coprime to n , Eve can factorize n by computing $\gcd(c_A, n) \neq 1$ (since not coprime) and $\neq n$ (because $c_A \neq 0$), so must be p or q . This allows Eve to obtain both p, q to compute d .

If coprime: Since $\gcd(e_A, e_B) = 1$, use Extended Euclid to compute u, v with $ue_A + ve_B = 1$. Then, compute $c_A^u \cdot c_B^v$, giving $c_A^u \cdot c_B^v \equiv (m^{e_A})^u \equiv (m^{e_B})^v \equiv m^{ue_A + ve_B} \equiv m^1 \equiv m$.

If negative u, v , use EEA to compute inverse, then proceed with inverse base and $|u|, |v|$.

9.2 Series 11

9.2.1 Let F finite field, $\mathcal{C} \subseteq F^n$ be a code with element-wise addition (linear).

Given $\min_{c \in \mathcal{C} \setminus \{0^n\}} \text{hw}(c) = 2t + 1$, prove $\mathcal{C}$ is t-error-correcting.

Let $c_1, c_2 \in \mathcal{C}, c_1 \neq c_2$. As $\mathcal{C}$ group, we know $c = c_1 - c_2 \in \mathcal{C}$. We have $d(c_1, c_2) = \text{hw}(c_1 - c_2) = \text{hw}(c) \geq 2t + 1$.

Justification: Two codewords differ at a position $\iff$ difference of their values at this position is nonzero. Also, $c_1 \neq c_2 \implies c \neq 0^n$.

Thus, we have $d_{\min}(\mathcal{C}) \geq 2t + 1$, so by Theorem 5.41, $\mathcal{C}$ is t-error-correcting.

Is it possible that for some $c \in \mathcal{C}$, up to $t + 1$ arbitrary errors can be corrected?

Let $c \in \mathcal{C}$ be arbitrary. Let $c_{\min} \in \mathcal{C} \setminus \{0^n\}$ such that its HD is $2t + 1$. We have $d(c, c + c_{\min}) = \text{hw}(c_{\min}) = 2t + 1$. So, c and $c + c_{\min}$ differ on exactly $2t + 1$ positions.

We can therefore change the first $t + 1$ positions of c to the ones of $c + c_{\min}$ to obtain a word w with $d(c, w) = t + 1$ and $d(c + c_{\min}, w) = t$. This word cannot be ECed.

As c was arbitrary, there is no codeword such that $t + 1$ arbitrary errors can be corrected.

9.2.1.1 Let $U \subseteq F^n, V \subseteq F^n$ be two linear codes.

Let $\mathcal{D} \subseteq F^{2n}$ be defined as $\mathcal{D} := \{(u \parallel (u + v)) \mid u \in U, v \in V\}$. Prove $d_{\min}(\mathcal{D}) = \min(2d_{\min}(U), d_{\min}(V))$.

$\leq$ **Direction:** For all $u \in U, (u \parallel u) \in \mathcal{D}$ since $0^n \in V$ because V linear. Similarly, for all $v \in V, (0^n \parallel v) \in \mathcal{D}$.

By choosing $u \in U \setminus \{0^n\}$ such that $\text{hw}(u) = \min_{c \in U \setminus \{0^n\}} \text{hw}(c)$ and $v \in V$ analogously, we get:

$$\begin{aligned} d_{\min}(\mathcal{D}) &= \min_{c \in \mathcal{D} \setminus \{0^n\}} \text{hw}(c) \leq \text{hw}(u \parallel u) = 2\text{hw}(u) \\ d_{\min}(\mathcal{D}) &= \min_{c \in \mathcal{D} \setminus \{0^n\}} \text{hw}(c) \leq \text{hw}(0^n \parallel v) = \text{hw}(v) \\ \text{Taking the minimum and again using the result } (d_{\min} &= \min \text{hw}) \text{ from below: } d_{\min}(\mathcal{D}) \leq \min(2d_{\min}(U), d_{\min}(V)) \\ \geq \textbf{Direction:} \text{ Let } u, v \text{ st } (u, v) &\neq (0^n \parallel 0^n). \text{ We have:} \\ \text{hw}(u \parallel (u + v)) &= \text{hw}(u) + \text{hw}(u + v) = \text{hw}(-u) + \text{hw}(u + v) \\ &\geq \text{hw}(-u + u + v) = \text{hw}(v) \end{aligned}$$

- If $v \neq 0^n, \text{hw}(v) \geq d_{\min}(V)$ by result below
- If $v = 0^n, \text{hw}(u \parallel u + v) = 2\text{hw}(u) \geq 2d_{\min}(U)$ by result below since $u \neq 0^n$ by assumption $(u, v) \neq (0^n \parallel 0^n)$.

Thus, by taking the minimum:

$$d_{\min}(\mathcal{D}) \geq \min(2d_{\min}(U), d_{\min}(V))$$

Conclusion $a(\geq \cap \leq)b \iff a = b$

9.2.2 Useful for linear codes over finite fields

- Hamming Weight satisfies the triangle inequality. $\forall x, y, \text{hw}(x + y) \leq \text{hw}(x) + \text{hw}(y)$. Requires proof.
- $d_{\min}(\mathcal{C}) = \min_{c \in \mathcal{C} \setminus \{0^n\}} \text{hw}(c)$. Requires proof. Show both inequalities for proof.